

Presupposition filtering in disjunction: What role does exclusive interpretation play?

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Intro. Kalomoiros and Schwarz (2024) found presupposition projection across disjunction is symmetric. This study aims to further investigate whether (and how) **presupposition (Ps)** projection across disjunctions is affected by exclusive disjunctive interpretations. In terms of theoretical predictions, Mayr and Romoli (2016) observed that in Strong Kleene semantics, exclusive disjunction yields uniform projection (i.e., no filtering for either disjunct). We made a novel observation that dynamic semantics generates the same prediction. To test whether this prediction (increasing exclusivity of disjunction *reduces* filtering) is borne out, we modified K&S (2024)’s paradigm and conducted a follow-up study in Mandarin (since we don’t expect significant cross-linguistic variations in this regard). Surprisingly, we found that increasing exclusivity of disjunction *enhances* filtering, contrary to theoretical predictions.

Motivation. In footnote 8, K&S noted that one reviewer raised a concern that the exclusive reading of disjunction might affect the interpretation of their results. This concern stems from K&S’s use of “either...or...” in both experiments, which sounds exclusive. That reviewer also pointed out that, according to Schlenker (2009)’s local context theory, an exclusive disjunction treats both disjuncts’ local contexts as the global context, thereby eliminating Ps filtering. Thus, while K&S interpreted the symmetry of Ps projection in their results as uniform *filtering* (i.e., Ps in either disjunct *can* be filtered), that reviewer worried that this symmetry could instead suggest uniform *projection* (Ps in either disjunct *cannot* be filtered). Although K&S argued against this uniform projection interpretation in footnote 24, we believe that the role of the exclusive reading of disjunction in its Ps projection warrants further theoretical and empirical investigations.

Theoretical prediction. Besides Schlenker’s local context theory, Strong Kleene semantics also predicts uniform projection for exclusive disjunction (Mayr & Romoli, 2016) (see Fig.1). We here made a novel observation that classic dynamic semantics predicts the same – the local contexts for both disjuncts of an exclusive disjunction is the global context.

$\forall$	1	0	#
1	0	1	#
0	1	0	#
#	#	#	#

Figure 1 Truth table of “xor”

(a) $C[\alpha][\neg\beta] \cup C[\beta][\neg\alpha]$	(e) $C - C[\alpha][\beta] - C[\neg\beta][\neg\alpha]$
(b) $C[\neg\alpha][\beta] \cup C[\neg\beta][\alpha]$	(f) $C - C[\neg\alpha][\neg\beta] - C[\beta][\alpha]$
(c) $C[\alpha][\neg\beta] \cup C[\neg\alpha][\beta]$	(g) $C - C[\alpha][\beta] - C[\neg\alpha][\neg\beta]$
(d) $C[\beta][\neg\alpha] \cup C[\neg\beta][\alpha]$	(h) $C - C[\beta][\alpha] - C[\neg\beta][\neg\alpha]$

Figure 2 Possible context change potentials for $C[\alpha \text{ xor } \beta]$

Fig.2 shows all eight possible ways to define the context change potential (CCP) of an exclusive disjunction $\alpha \text{ xor } \beta$. Turning to their definedness conditions, for (a), $C[\alpha]$ and $C[\beta]$ need to be defined, meaning there is no filtering for α and β . For (e), $C[\alpha]$ and $C[\neg\beta]$ need to be defined. Since $C[\neg\beta] = C - C[\beta]$, this means $C[\beta]$ should also be defined, equating the situation with (a). (b) and (f) are also similar to (a). For (c), $C[\alpha]$, $C[\alpha][\neg\beta]$, and $C[\neg\alpha][\beta]$ should be defined. Since the definedness of $C[\alpha][\neg\beta]$ requires the definedness of $C[\alpha][\beta]$, this means β should be defined on both $C[\alpha]$ and $C[\neg\alpha]$, i.e., the entire context set C . Thus, there is no filtering for both α and β in (c). The cases for (g), (d), and (h) are similar to (c).

Thus, classic trivalent logics, dynamic semantics, and Schlenker’s local context theory all predict no filtering for an exclusive disjunction. In other words, they all predict that increasing exclusivity of disjunction *reduces* filtering *uniformly* (for Ps in both disjuncts).

Design. We conducted a study in Mandarin (N = 197) to examine whether the exclusivity of disjunction affects its Ps filtering, and if so, whether the above theoretical prediction is borne out. We chose Mandarin because it is a sample of convenience and we do not expect significant cross-linguistic differences in this effect.

To manipulate the exclusive interpretation of disjunction, we introduced two cross-subject factors: **disjunction type (DisjType)**: “huozhe” vs. “yaome...yaome...” and environmental **monotonicity (MonoType)**: downward-entailing(DE) vs.upward-entailing(UE)). We expected “yaome yaome” to be more exclusive than “huozhe” and disjunctions in a UE environment to

be more exclusive than in DE ones due to exclusive implicatures. These expectations were confirmed in a norming task (see (1) for an example trial and Table 1 for results).

As for the within-subject conditions, we adopted K&S (2024)'s design. They crossed **PresuppositionType** (**PsType**: Ps vs. NoPs) with **Order** (First vs. Second), yielding four conditions: PsFirst, PsSecond, NoPsFirst, and NoPsSecond. "PsFirst" means the first disjunct contains the Ps trigger; "NoPsFirst" replaces that Ps trigger in the first disjunct with a suitable non-presuppositional predicate. "PsSecond" and "NoPsSecond" means the Ps trigger (or its NoPs counterpart) is in the second disjunct. To prevent global accommodation, test sentences in these four conditions are all preceded by a context that is **explicitly ignorant (EI)** about the relevant Ps. K&S also included two additional control conditions: EISimplePs, and SSimplePs (the first "S" in "SSimplePs" means the context is **supportive** of the relevant Ps). The difference of ratings between them establishes the cost of local accommodation.

Stimuli. We selected two effective Ps triggers, "jie" (quit) and "zhidao" (know), based on pilot data showing ratings for these in the EISimplePs condition were significantly lower than in SSimplePs. In our main experiment, we created 2 items for each trigger. Each participant completed 15 trials in total: 8 critical trials (2 conditions per item), 4 catch trials, 2 filler trials, and 1 inferential task. A critical trial consisted of: (i) an either EI or S context; (ii) a test sentence involving a bathroom disjunction; (iii) a judgment task for the naturalness of the test sentence in the context on a 7-point scale. See (2) for an example trial.

Predictions. (a) If the exclusive interpretation of disjunction affects its Ps filtering, we expect significant two-way interactions for DisjType*PsType and MonoType*PsType.

(b) If that effect is *uniform* for filtering in both disjuncts, we expect no significant three-way interactions including Order (i.e., Order*DisjType*PsType and Order*MonoType*PsType).

(c) If increasing exclusivity *reduces* filtering *uniformly*, we expect (a) and (b) to hold, and that ratings in the Ps conditions should *decrease relative to* the NoPs conditions, when either DisjType changes from "huozhe" to "yaome yaome" or MonoType changes from DE to UE.

Participants. We recruited 493 native Mandarin speakers. 296 of them were excluded for failing at least one of the four catch trials. The remaining 197 responses were analyzed.

Results. We analyzed the data with R's *clmm* function, using maximal random effects for participants. Our analyses revealed significant two-way interactions for DisjType*PsType ($\beta=0.635$, $SE=0.314$, $p=0.043$) and for MonoType*PsType ($\beta=0.633$, $SE=0.298$, $p=0.034$), while neither of the three-way interactions involving Order (Order*DisjType*PsType and Order*MonoType*PsType) reached significance. Unexpectedly, interaction plots (see Fig.3 and Fig.4) show that ratings in the Ps conditions actually *increase relative to* the NoPs conditions, when either DisjType changes from "huozhe" to "yaome yaome" or MonoType changes from DE to UE. (Notably, in the "yaome yaome" or UE conditions, ratings in the Ps conditions even exceeded ratings in NoPs conditions. This preference may reflect the *Maximize Presupposition* principle when Ps is felicitous.) See Fig.5 for aggregated results.

Discussion. Predictions (a) and (b) hold, but (c) does not. Thus, our results suggest increasing exclusivity of disjunction *enhances* filtering *uniformly*, contrary to the theoretical prediction.

Conclusion. We first examined theoretical predictions for how exclusivity of disjunction might affect its presupposition filtering. Schlenker's local context theory, trivalent logics, and dynamic semantics (which we observed) all predict increasing exclusivity reduces filtering uniformly. To test this prediction, we modified K&S (2024)'s experimental paradigm. Our results indicate that: (1) exclusivity of disjunction does affect its presupposition filtering, suggesting that the computation of presuppositions is affected by scalar implicatures; (2) this effect is uniform for filtering in both disjuncts; (3) the direction of such effect is opposite to theoretical predictions: increasing exclusivity actually enhances filtering uniformly. This unexpected pattern calls for more empirical validation. If proven to be robust cross-linguistically, it may have profound implications for theories of presupposition projection.

(1) (Translation of) [A trial](#) (from **the norming task**) of “huozhe” in a UE environment:

Li said: “I believe Zhang or (huozhe) Wang will come to the party.”
 In fact, both Zhang and Wang came to the party.
 Do you consider Li’s prediction correct or incorrect?

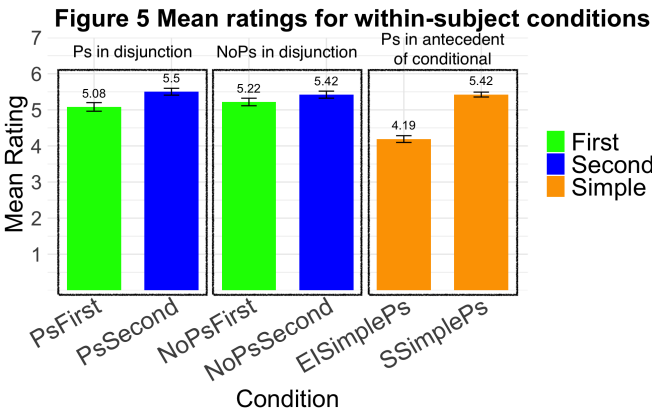
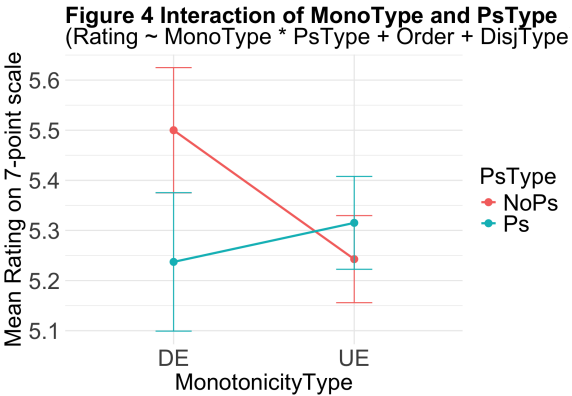
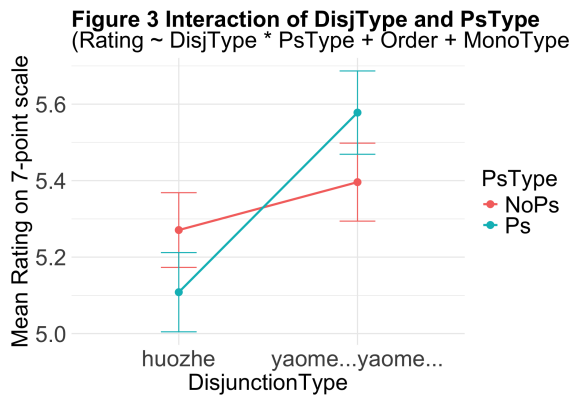
[Choosing “incorrect” will be analyzed as an “exclusive reading” of disjunction]

Table 1 Percentage of exclusive reading of disjunction across participants in the norming task

	DE environment	UE environment
...huozhe...	0%	23.3%
Yaome...yaome...	36.7%	53.3%

(2) [A trial](#) (from **the main experiment**) of **PsFirst** within-subject condition + “**huozhe**” in **a DE environment** cross-subject condition, with “**jie**” (**quit**) as the Ps trigger:

我之前完全不认识小李，不知道他有没有喝过酒。在今晚的饭局上，大多数人都喝酒了，但小李滴酒不沾，于是我想：
如果小李已经戒酒了或者从来不喝酒，那么他今晚的行为很合理。
 请给划线句子在语境中的自然程度打分：
 I didn’t know Li at all before, and I didn’t know if he ever drank. At tonight’s dinner party, most people drank, but Li didn’t drink a drop, so I thought:
If Li has quit drinking or (huozhe) never drinks, then his behaviour tonight makes sense.
 Please rate how natural the underlined sentence is in the context:
 [7-point scale; ends of scale: completely unnatural 1 – completely natural 7]



Selected References:

Kalomoiros, A., & Schwarz, F. (2024) Presupposition projection from “and” vs “or”: Experimental data and theoretical implications. *Journal of Semantics*, 41(3-4), 331-72.
Mayr, C., & Romoli, J. (2016) Satisfied or exhausted: An ambiguity account of the Proviso Problem. *Proceedings of SALT 26*, 892-912.
Schlenker, P. (2009). Local contexts. *Semantics and Pragmatics*, 2(3), 1–78.